Ahmadreza Tahmasebi

Ph.D. researcher in Mechanical Engineering, Tarbiat Modares University

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A RESEARCH PROFILE

I develop additive manufacturing processes and test what the printed parts can carry. My doctoral research is on volumetric additive manufacturing, including hybrid injection–volumetric printing of opaque, filled and low-viscosity resins. My master's work produced a biodegradable surgical mesh by direct ink writing with continuous PGA fibre embedded during printing. Interests: additive manufacturing, volumetric AM, mechanical metamaterials, composites, biomechanics, and machine learning in AM.

B EDUCATION

Sep 2026 – present
Ph.D., Mechanical Engineering
Tarbiat Modares University, Tehran · Thesis: volumetric additive manufacturing · Advisor: Dr. Amirhossein Behravesh

2023 – 2026
M.Sc., Mechanical Engineering
Tarbiat Modares University, Tehran · Thesis: *Fabrication of a surgical mesh reinforced with continuous polyglycolic acid fibres by direct ink writing, and evaluation of its mechanical and biological properties* · Advisor: Dr. Amirhossein Behravesh

2020 – 2023
B.Sc., Automotive Mechanics Technology
Islamic Azad University, Yadegar Emam branch · Thesis: *Design of an 18-metre telescopic conveyor belt for unloading truck trailers* · Advisor: Dr. Amir Khorram

C PUBLICATIONS

[6] O. Kordi, A. H. Behravesh, M. M. Ghafari, A. Kavian, **A. R. Tahmasebi**, A. A. Milani, G. Rizvi. **Hybrid injection–volumetric additive manufacturing for rapid fabrication of multi-material and encapsulated reinforced structures in low-viscosity resins.** *Progress in Additive Manufacturing* 11(8), 7139–7160, 2026. doi:10.1007/s40964-026-01720-w

[5] H. Amirahmadi, A. H. Behravesh, A. Amirahmadi, I. Ghaderi, S. K. Hedayati, **A. Tahmasebi**, A. Bakhtiary. **Screw extrusion-based additive manufacturing of fiber-reinforced high-temperature thermoplastics.** *Iranian Polymer Journal* 35(3), 483–493, 2026. doi:10.1007/s13726-025-01536-5

[4] M. Hematibahar, M. Kharun, N. A. Stashevskaya, A. A. Milani, **A. Tahmasebi**, O. Kordi, A. Rossi, C. H. Martins. **Evaluation of mechanical properties of 3D printed PLA-CF reinforced concrete under Atlantic coastal conditions of Brazil.** *Discover Civil Engineering* 2(1), 208, 2025.

[3] M. M. Ghaffari, O. Kordi, M. A. Yusefi, **A. R. Tahmasebi**. **Effect of applying filters on the visual accuracy of parts produced by volumetric additive manufacturing.** *Iranian Journal of Manufacturing Engineering* 12(9), 13–24, 2025. doi:10.22034/ijme.2025.529713.2094

[2] M. Hematibahar, R. S. Fediuk, N. I. Vatin, A. Milani, **A. Tahmasebi**, O. Kordi, M. Kharun, G. R. Fediuk, A. O. Shangutov, Y. K. Gitman. **Concrete column performance enhanced by 3D-printed honeycomb, chiral auxetic, and re-entrant lattices via FDM and DLP methods.** *Materials Physics and Mechanics* 53(5), 164–180, 2025. doi:10.18149/MPM.5352025_14

[1] H. Amirahmadi, A. H. Behravesh, A. Amirahmadi, I. Ghaderi, S. K. Hedayati, **A. Tahmasebi**, A. Bakhtiyari. **Screw extrusion-based additive manufacturing of polycarbonate, PEEK and continuous fiber–reinforced PEEK composites.** Preprints.org, 2024. doi:10.20944/preprints202401.1943.v1

Under review
Journal manuscript on the PGA-fibre-reinforced DIW surgical mesh, submitted to *Biomedical Materials & Devices*.

D RESEARCH EXPERIENCE

2023 – present
Graduate researcher, additive manufacturing
Tarbiat Modares University. Adapted a DIW printer for continuous-fibre printing; volumetric and hybrid injection–volumetric printing; mechanical, rheological, chemical and in-vitro testing.

Mar 2022 – Jan 2023	**Print pattern optimisation in additive manufacturing** Research project
Nov 2022 – Feb 2023	**Portable dry-mix batching plant** Islamic Azad University, Yadegar Emam branch · Advisor: Dr. Khorram
Sep – Nov 2022	**Medical chair design** Islamic Azad University, Yadegar Emam branch · Advisor: Dr. Khorram
Apr 2021 – Aug 2022	**Research assistant, vertical wind turbine design and construction** Islamic Azad University, Yadegar Emam branch · Supervisor: Dr. Shakoori
Sep 2020 – Mar 2021	**Cushion behaviour analysis, Rah Neshan competition** Supervised by the Vice Presidency for Science and Technology, National Elite Foundation · top-ten team
Jun 2018 – Dec 2019	**Research assistant, telescope mount deflection analysis** Institute for Research in Fundamental Sciences (IPM), Iranian National Observatory R&D team · Team leader: Mr. Surena Fatemi · Advisor: Dr. Khosroshahi

E TEACHING

Apr – Jul 2022	**Teaching assistant, Machining Workshop** Islamic Azad University, Yadegar Emam branch · Dr. Nouri
Jan – May 2022	**Teaching assistant, Vehicle Dynamic Vibration** Islamic Azad University, Yadegar Emam branch · Dr. Khorram

F INDUSTRY & SERVICE

Industry	**Engineer, APPACO** Automotive industry
Industry	**SAPCO** Automotive industry
Jun – Jul 2024	**Executive committee, 7th International Conference on Nondestructive Testing** Iranian Society of Nondestructive Testing (IRNDT) · mechanical engineer
2022, 2024	**Executive committee, 12th and 13th International Conferences on Internal Combustion Engines and Petroleum** Iran Khodro Powertrain Company (IPCO) · mechanical engineer
Apr – Nov 2017	**Race engine mechanic** Motorcycle and Automobile Federation of Iran (MAFIRI) · tuned and upgraded engines for competition

G SKILLS & TRAINING

DESIGN	SolidWorks, CATIA, Shapr3D, AutoCAD, GD&T
SIMULATION	Abaqus, ADAMS, COMSOL Multiphysics, SolidWorks Simulation
COMPUTING	MATLAB, Python, machine learning, LaTeX
LAB	DIW, FDM, DLP, volumetric AM; tensile and burst testing, rheometry, ATR-FTIR, SEM, MTT
CERTIFICATES	Machine Learning 2024 · GD&T 2023 · CATIA 2023 · Python 2022 · MATLAB (Univ. of Tehran) 2021 · SolidWorks 2021
MEMBERSHIPS	Iranian Society of Nondestructive Testing · Society of Manufacturing Engineering of Iran · Researchers and Elites of Islamic Azad University
LANGUAGES	Persian (native), English (fluent)